

杭州电子科技大学学生考试卷 (A) 卷

考试课程	通信电路与系统		考试日期	年 月 日		成绩	
课程号	A040095s	教师号		任课教师姓名		陈瑾、林弥、周涛、 苏江涛、邓天松、孙 朋飞、张伟、刘艳	
考生姓名		学号 (8 位)		年级		专业	

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
得分						

I. Fill-in-the-Blanks Exercise (3' each)

1. The frequency range of voice signals is 300-3400 Hz with the propagation speed of $c = 3 \times 10^8$ m/s in free space. So the wavelength range of the voice signals can be estimated as 8.82×10^4 m- 1×10^6 m.

2. For the LC parallel resonant circuit, as shown in Fig.1, under resonant condition, the quality factor is

Q_0 , and the magnitude of capacitor current is $Q_0 |I_s(j\omega)|$.

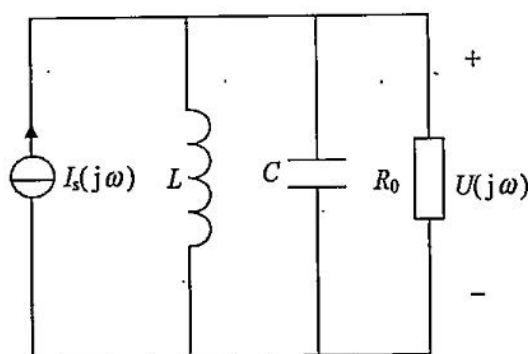


Fig.1 LC parallel resonant circuit

3. For class C power amplifier, its virtual quiescent point is at the cutoff state because the biasing voltage of the base-emitter PN junction is lower than the threshold voltage U_{th} (i.e. reversely biased). Therefore, you can obtain that the quiescent current I_{CQ} is 0. (0, >0 or <0? Please choose your answer and fill in the blank)

4. For a PA, P_{out} means output power, P_{in} means input power, P_{DC} means the supplied DC power and

PAE or η_{add} means power-added efficiency. For PAE, based on P_{out} , P_{in} , and P_{DC} , it can be given as

$$\text{PAE} = \frac{P_{\text{out}} - P_{\text{in}}}{P_{\text{DC}}}$$

5. The two figures, as shown in Fig.2, are the basic composition of the capacitance feedback and inductance feedback LC oscillators. Fig.2 (a) is called 考必兹电路 and Fig.2 (b) is called 哈特莱电路.

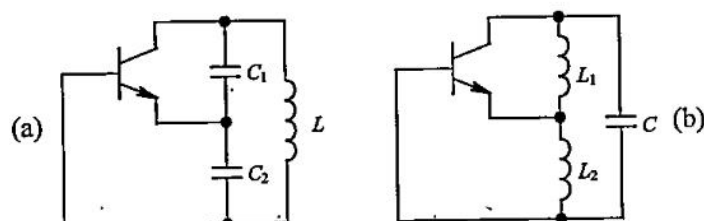


Fig.2 Two basic types of feedback LC oscillator

6. Fig.3 is reactance characteristic of quartz crystal. When $f_s < f < f_p$, X is positive. Is the circuit inductive or capacitive? inductive (感性).

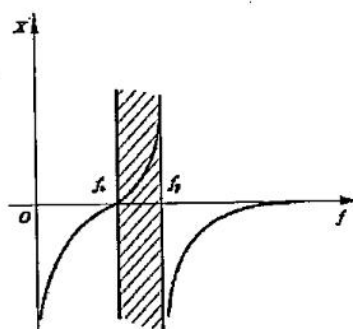


Fig.3 Reactance characteristic of quartz crystal

7. There are three basic and classic modulation methods, which are frequency modulation and phase modulation and 幅度调制.
8. For large-signal diode envelope detector, there are two kinds of common distortions, which are diagonal clipping and 负峰切割失真.
9. For AM, DSB, and SSB, which amplitude modulation mode has the highest transmission efficiency? SSB.
10. As we know, energy or power of a wave is closely related to its voltage amplitude. From the view of power consumption, for FM and AM, which one is just the power redistribution of the previous

carrier wave, without changing the total power after modulation? FM.

II. Analysis and Calculation

1. (10') Fig.4 is the diagram of the AM transmitting system. According to Fig.4, please answer the following questions:

- (1) Complete the transmitting system in Fig.4.
- (2) Point out the characteristics of the signals corresponding to point A, point B or point C in Fig.4, and illustrate their relationship.
- (3) To reduce the power consumption of PA- I in Fig.4, how to do?

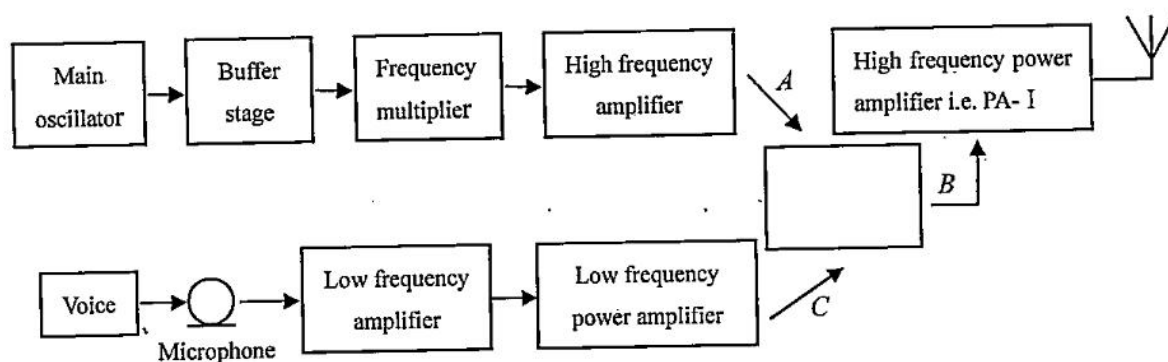


Fig.4 Diagram of the AM transmitting system

(1) Modulator (注: 写中文“调制器”也得分) (1')

(2) A 点对应的是高频载波信号 (1'), B 点对应的是调幅波 (1'), c 对应的是低频调制信号 (1'); 调幅波是由调制信号对载波信号的幅度调制后所得的波 (1')。

(3) 为了降低 PA- I 的功耗可采取如下措施: ①选择合适的静态工作点, 从降低偏置电路功耗的角度降低 PA- I 的功耗; ②选择功耗低的晶体管作为 PA- I 的管芯 (注: 可选择具有低功耗材料跟器件结构的晶体管); ③抑制谐波, 降低谐波功耗。注: 该题具有一定的开放性, 回答出上述一种措施或上述措施之外的其它合理措施即可。 (5')

2. (10') The impedance characteristic curve and bandwidth of a LC resonant circuit without load are shown in Fig.5 (a) and Fig.5 (b), respectively. According to Fig.5, please answer the following questions:

(1) From Fig.5 (a), it can be inferred that the LC resonant circuit here is parallel resonant circuit. Please write down the reason.

(2) Assuming the lossless capacitance C is 100 pF in total for this LC resonant circuit, please calculate the values of the corresponding lossless inductance L , and quality factor Q_0 under the resonant condition combining with Fig.5 (b).

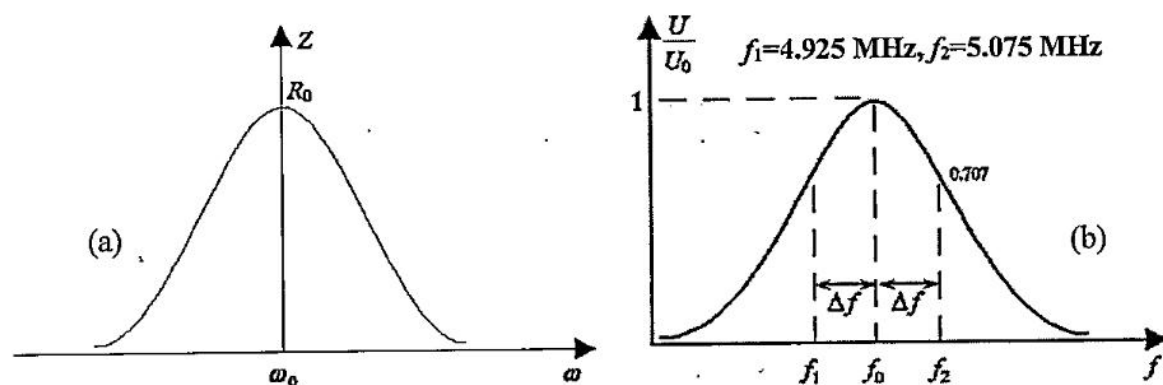


Fig.5 (a) impedance characteristic curve and (b) bandwidth of a LC resonant circuit;

(1) Fig.5 (a) 是该题所述的空载 LC 谐振回路的阻抗特性，该图表明此谐振回路的阻抗最大值出现在谐振频率 ω_0 上；这一结果符合 LC 并联谐振回路的特性，因此可推知该谐振回路为并联型谐振回路(2')。

(2) 根据题意知，已知空载 LC 并联谐振回路中总的无损电容 $C=100$ pF，结合 Fig.5 (b) 求谐振时对应的电感 L 和品质因数 Q_0

① L 的求解

由 Fig.5 (b) 得：

$$\Delta f = (f_2 - f_1) / 2 = (5.075 \text{ MHz} - 4.925 \text{ MHz}) / 2 = 0.075 \text{ MHz} \quad (2') \quad (\text{这里算带宽的也给分})$$

$$f_0 = f_1 + \Delta f = 4.925 \text{ MHz} + 0.075 \text{ MHz} = 5 \text{ MHz} \quad (2')$$

又题目已知： $C=100$ pF；

$$L = \frac{1}{(2\pi f_0)^2 C} = \frac{1}{(2 \times 3.14 \times 5 \times 10^6 \text{ Hz})^2 \times 100 \times 10^{-12} \text{ F}} \approx 10.14 \mu\text{H} \quad (2')$$

② Q_0 的求解

$$Q_0 = \frac{f_0}{2\Delta f} = \frac{5 \times 10^6 \text{ Hz}}{2 \times 0.075 \times 10^6 \text{ Hz}} \approx 33.33 \quad (2')$$

3. (12') A resonant power amplifier, assuming $E_c = 24 \text{ V}$, output power $P_o = 5 \text{ W}$, please answer the following questions:

(1) When collector efficiency η_c is 60%, please calculate the transistor dissipate power P_c and the DC current I_{c0} ,

(2) If the value of the P_o holding 5 W, when the value of the η_c rises from 60% to 80%, how much does the value of the P_c change?

(1) 由题目知: $\eta_c = 60\%$, $P_o = 5 \text{ W}$;

又 $\eta_c = \frac{P_o}{P_{DC}}$, 其中 P_{DC} 为电源的直流功率;

$$\therefore P_{DC} = \frac{P_o}{\eta_c} = 5\text{W} / 60\% = 8.33\text{W} \quad (3')$$

$$\therefore P_c = P_{DC} - P_o = 8.33\text{W} - 5\text{W} = 3.33\text{W} \quad (3')$$

$$I_{c0} = \frac{P_{DC}}{E_c} = \frac{8.33\text{W}}{24\text{V}} = 0.35\text{A} \quad (2')$$

(2) 如果 P_o 保持 5 W, η_c 由 60% 提高到 80% 时, P_c 的改变量计算如下:

设 η_c 为 60% 和 80% 时, 对应的晶体管损耗功率分别为 $P_{c80\%}$, $P_{c60\%}$;

$$P_c = P_{DC} - P_o = P_o / \eta_c - P_o$$

$$\begin{aligned} P_{c80\%} - P_{c60\%} &= (P_o / 80\% - P_o) - (P_o / 60\% - P_o) \\ &= P_o / 80\% - P_o / 60\% = 5\text{W} / 80\% - 5\text{W} / 60\% \\ &= -2.08\text{W} \end{aligned} \quad (4')$$

4. (10') Fig. 6 is the AC equivalent circuit of an LC oscillator. The resonant frequencies of the three parallel LC loops are:

$$f_1 = \frac{1}{2\pi\sqrt{L_1C_1}}, \quad f_2 = \frac{1}{2\pi\sqrt{L_2C_2}}, \quad \text{and} \quad f_3 = \frac{1}{2\pi\sqrt{L_3C_3}}.$$

(1) If the oscillator in Fig.6 is the Colpitts oscillator, please find out the quantitative relation among f_1 , f_2 , f_3 , and the oscillation frequency f_{osc} .

(2) If the oscillator in Fig.6 is the Hartley oscillator, please find out the quantitative relation among f_1 , f_2 , f_3 , and the oscillation frequency f_{osc} .

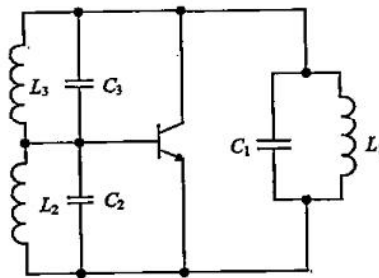


Fig.6 AC equivalent circuit of an LC oscillator

(1) 根据考必兹振荡器的特点, 在 f_{osc} 处, 与晶体管发射极相连的 L_1C_1 回路以及 L_2C_2 回路需呈容性, 即 f_{osc} 应大于 f_1 和 f_2 ; 基-集极相连的 L_3C_3 回路需呈感性, 即 f_{osc} 应小于 f_3 ; 所以 $f_1 < f_{osc} < f_3, f_2 < f_{osc} < f_3$ 。 (5')

(2) 根据哈特莱振荡器的特点, 在 f_{osc} 处, 与晶体管发射极相连的 L_1C_1 回路以及 L_2C_2 回路需呈感性, 即 f_{osc} 应小于 f_1 和 f_2 ; 基-集极相连的 L_3C_3 回路需呈容性, 即 f_{osc} 应大于 f_3 ; 所以 $f_1 > f_{osc} > f_3, f_2 > f_{osc} > f_3$ 。 (5')

5. (10') Fig.7 shows the realization of $u_o(t)$. In Fig.7, $u_\Omega(t) = U_{\Omega m} \cos(\Omega t)$ and $u_c(t) = U_{cm} \cos(\omega_c t)$ are modulation signal and high-frequency carrier, respectively. According to Fig.7, please answer the following questions:

(1) If $u_o(t)$ is DSB wave, please draw the waves of $u_\Omega(t)$, $u_c(t)$ and $u_o(t)$ briefly, respectively.

(2) If $u_o(t)$ is DSB wave, please draw the frequency spectrum of $u_o(t)$?

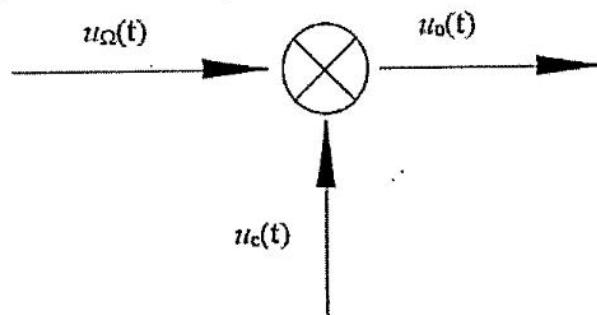
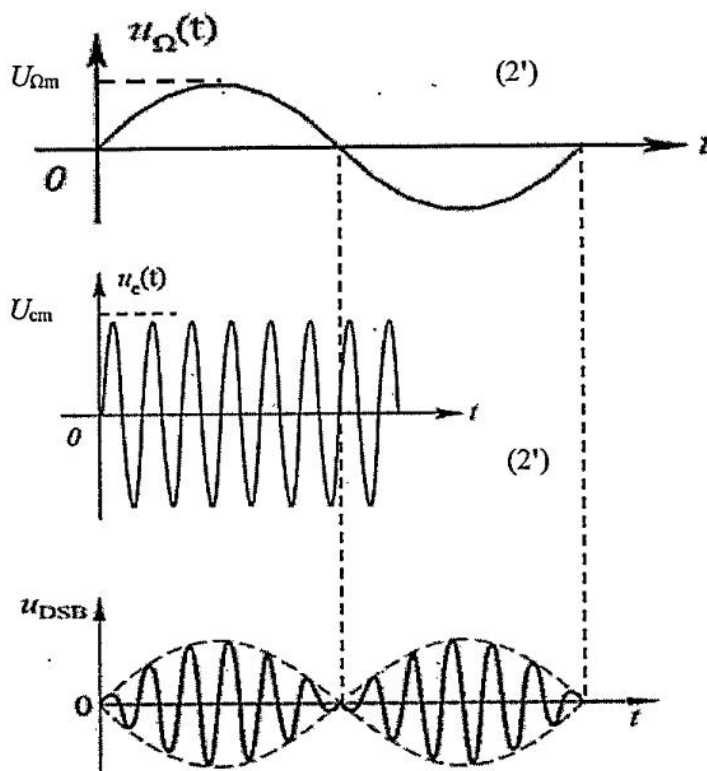


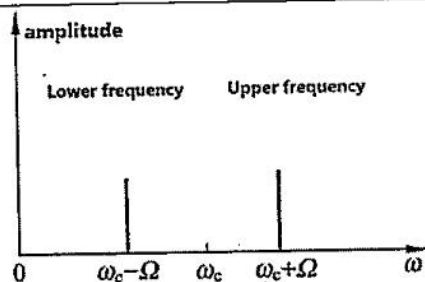
Fig.7 Realization of $u_o(t)$

(1)

画图题不唯一，大家可酌情给分，调制信号、载波和DSB波一致即可。不一致的请酌情给分。



(2)



(3')

6. (10') The circuit of large signal peak envelope detector is shown in Fig.8, given the modulation frequency of the amplitude-modulated wave i.e. F is 50 Hz, $R_L = 5 \text{ k}\Omega$, modulation factor $m_a = 0.6$. In order to avoid diagonal distortion, please find the maximum capacitor C .

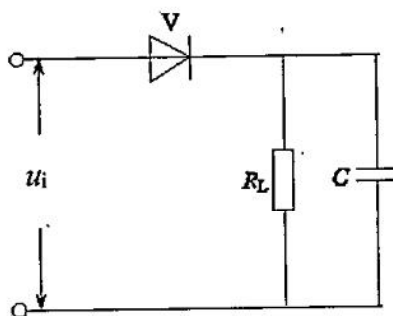


Fig.8 The circuit of large signal peak envelope detector

解：不产生对角线失真的条件为：电容放电的速度大于等于输入信号包络下降的速度。 (2')
因此为了不产生对角线失真， C 应满足的条件为：

$$C \leq \frac{\sqrt{1-m_a^2}}{R_L m_a \Omega} \quad (4')$$

又由题目知 $R_L = 5 \text{ k}\Omega$, $m_a = 0.6$, $F = 50 \text{ Hz}$

$\therefore$

$$C \leq \frac{\sqrt{1-m_a^2}}{R_L m_a \Omega} = \frac{\sqrt{1-m_a^2}}{R_L m_a 2\pi F} = \frac{\sqrt{1-0.6^2}}{5 \times 10^3 \times 0.6 \times 2 \times 3.14 \times 50} = 0.85 \mu\text{F} \quad (4')$$

所以 C 最大取 $0.85 \mu\text{F}$ 。

7. (8') Assuming the FM signal $u_{FM}(t) = 5 \cos(2\pi \times 10^7 t + 175 \sin 2\pi F t)$, the modulate signal $u_{\Omega}(t) = 2\pi \cos 2\pi F t$, and the maximum frequency shift $\Delta f_{\max} = 70 \text{ kHz}$, please:

- (1) Write down the value of the carrier wave frequency f_c and the maximum phase shift m_f ,
- (2) Calculate the modulate signal frequency F and the effective bandwidth B_{FM} .

解:

(1). 根据 $u_{FM}(t) = 5 \cos(2\pi \times 10^7 t + 175 \sin 2\pi F t) = U_{cm} \cos(\omega_c t + m_f \sin \Omega t)$, 知:

$$m_f = 175 \quad (2')$$

$$f_c = 10 \text{ MHz} \quad (2')$$

$$(2) \text{ 又 } m_f = \frac{\Delta f_{\max}}{F}, \therefore F = \frac{\Delta f_{\max}}{m_f} = \frac{70 \times 10^3 \text{ Hz}}{175} = 400 \text{ Hz} \quad (2')$$

$$B_{FM} = 2(m_f + 1)F = 2 \times (175 + 1) \times 400 \text{ Hz} = 140.8 \text{ kHz} \quad (2')$$

杭州电子科技大学学生考试卷（B）卷

考试课程	通信电路与系统		考试日期	2021 年 月 日		成绩	
课程号	A040095s	教师号		任课教师姓名		陈瑾、林弥、周涛、 苏江涛、邓天松、孙 朋飞、张伟、刘艳	
考生姓名		学号（8位）		年级		专业	

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I. Judgment questions (2' each)

1. LC series resonant circuit and LC parallel resonant circuit both have frequency-selective function. (√)
2. When the operating frequency is greater than the resonant frequency, the LC series resonant circuit is capacitive. (√)
3. The higher the current conduction angle, the higher the efficiency. (x)
4. The efficiency of the power amplifier reaches the maximum in the over-voltage status. (x)
5. Hartley circuit has higher oscillation frequency than Colpitts circuit. (x)
6. Quartz crystal has high frequency stability due to its high quality factor. (√)
7. Amplitude modulation circuit is a linear spectrum shifting circuit. (√)
8. SSB modulation mode has the highest transmission efficiency. (√)
9. Angle modulation includes frequency modulation and phase modulation. (√)
10. Theoretically, both FM and PM signals have infinite spectrum structure. (√)

II. Analysis and Calculation

1. (15') Fig.1 is compositions diagram of super heterodyne receiving system. According to Fig.1, please answer the following questions:

- (1) Complete the super heterodyne receiving system in Fig.1.
- (2) Draw the waveform of point A and point B in Fig.1.
- (3) For a super heterodyne receiving system, in order to minimize the effect on environment, what method can be adopted?

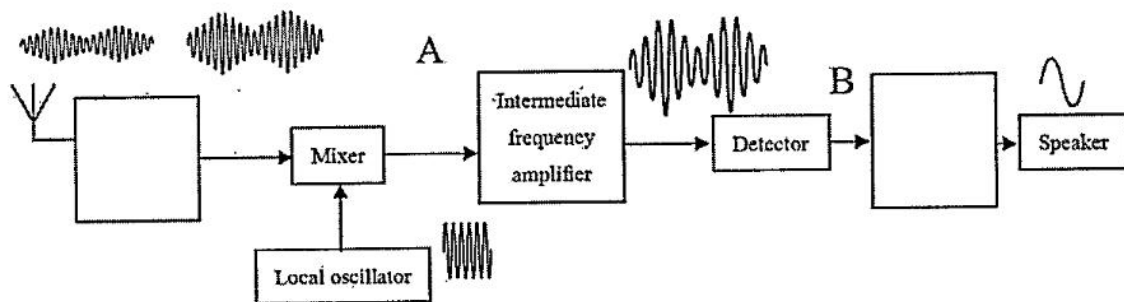


Fig.1 Compositions Diagram of Super Heterodyne Receiving System

(1) 高频放大器 (High frequency amplifier) (2'), 低频放大器 (Low frequency amplifier) (2')

(2) A (3') B (3')



(3) 可就接收系统中的各模块, 尽量提高效率, 减小功耗 (回答合理措施即可)。(5')

2. (10') In a LC parallel resonant circuit, assuming $L = 10 \mu\text{H}$, bandwidth $B_{0.7} = 100 \text{ kHz}$, and resonance frequency $f_0 = 1.5 \text{ MHz}$, please answer the following questions:

- (1) Calculate the values of Q_0 and C for this LC parallel resonant circuit;
- (2) Calculate the value of the parallel resistance which can make the value of $B_{0.7}$ alter from 100 kHz to 500 kHz .

(1) 由已知条件得:

$$C = \frac{1}{(2\pi f_0)^2 L} = \frac{1}{(2 \times 3.14 \times 1.5 \times 10^6 \text{ Hz})^2 \times 10 \times 10^{-6} \text{ H}} \approx 1.13 \text{ nF} \quad (3')$$

$$Q_0 = \frac{f_0}{B_{0.7}} = \frac{1.5 \times 10^6 \text{ Hz}}{100 \times 10^3 \text{ Hz}} = 15 \quad (3')$$

$$(2) Q_0 = \frac{f_0}{B_{0.7}} = \frac{1.5 \times 10^6 \text{ Hz}}{500 \times 10^3 \text{ Hz}} = 3 \quad (2')$$

$$R_0 = Q_0 \omega_0 L = 3 \times 2 \times 3.14 \times 1.5 \times 10^6 \times 10 \times 10^{-6} \Omega = 282.6 \Omega \quad (2')$$

(注: 每个单位没写或不对扣 1 分, 下同)

3. (12') A resonant power amplifier with operating at the critical point state. Assuming $E_c = 16 \text{ V}$, current conduction angle (CCA) $\theta = 55^\circ$, $\alpha_0(55^\circ) = 0.2005$, $\alpha_1(55^\circ) = 0.3658$, collector current pulse maximum value $I_{cmax} = 2 \text{ A}$, collector voltage utilization factor $\xi = 0.9$.

(1) Please calculate output power P_o , and voltage source providing power P_s ;

(2) Please calculate the collector efficiency η_c and collector equivalent load resistor R_c .

$$(1) \quad U_{c1m} = \xi \times E_c = 0.9 \times 16 \text{ V} = 14.4 \text{ V} \quad (1')$$

$$I_{c1m} = \alpha_1(\theta) \times I_{cmax} = 0.3658 \times 2 \text{ A} = 0.73 \text{ A} \quad (1')$$

$$P_o = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 14.4 \times 0.73 \text{ W} = 5.26 \text{ W} \quad (2')$$

$$I_{c0} = \alpha_0(\theta) \times I_{cmax} = 0.2005 \times 2 \text{ A} = 0.4 \text{ A} \quad (1')$$

$$P_s = E_c \times I_{c0} = 16 \times 0.4 \text{ W} = 6.4 \text{ W} \quad (2')$$

$$(2) \quad \eta_c = \frac{P_o}{P_s} = \frac{5.26}{6.4} = 82.2\% \quad (3')$$

$$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{14.4}{0.73} \Omega = 19.7 \Omega \quad (2')$$

4. (12') The AC equivalent circuit of Clapp oscillator is shown in Fig.2. According to Fig.2, please answer the following questions:

- (1) assuming the value of $L=50\ \mu\text{H}$, $C_1=C_2=2000\ \text{pF}$, $C=100\sim150\ \text{pF}$, calculate the resonant frequency f_0 .
- (2) Give the advantages and disadvantages of the Clapp oscillator.
- (3) To raise the frequency stability, based on Clapp oscillator, the Seiler oscillator can be obtained. Please draw the AC equivalent circuit of the Seiler oscillator;

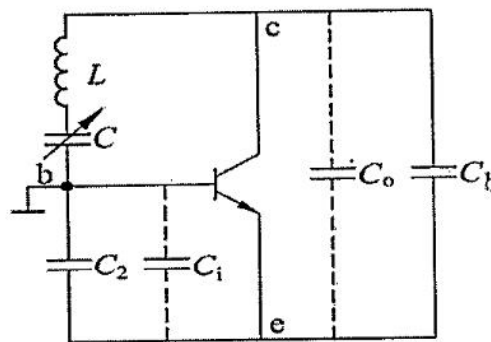


Fig.2 AC equivalent circuit of Clapp oscillator

$$(1) f_0 \approx \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{50 \times 10^{-6} \times (100 \sim 150) \times 10^{-12}}} \text{ Hz} = 1.84 \sim 2.25 \text{ MHz} \quad (3')$$

(2) Advantages: (2')

Good waveform;

High frequency stability;

High oscillation frequency;

Minimize the effect of transistor's C_0 and C_i on frequency; improve frequency stability further.

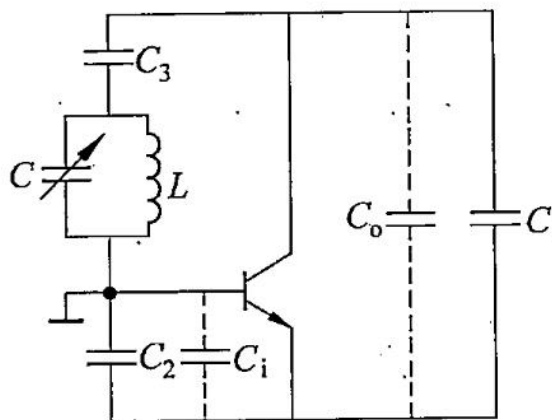
Disadvantages: (2')

If C_1 and C_2 are too large oscillation amplitude is too small;

When tuning C to increase f , oscillation amplitude is significantly decreased and even fail to oscillate;

Output amplitude is not stable in the band range, so the frequency coverage coefficient is not large, about 1.2 to 1.3.

(3) Seiler 等效电路如下图所示: (5')



5. (12') A modulated signal is $u(t)=0.7\cos 4950\times 10^3\pi t+0.7\cos 5050\times 10^3\pi t(\text{V})$,

(1) What kind of modulated wave it is? Please rewrite the expression with obvious information about carrier wave frequency and voice signal frequency (standard form);

(2) If $U_{cm}=2\text{ V}$, Calculate its total power with unit resistance ($R_L=1$) and the frequency bandwidth B .

(3) Draw its waveform and frequency spectrum;

(1) DSB (2')

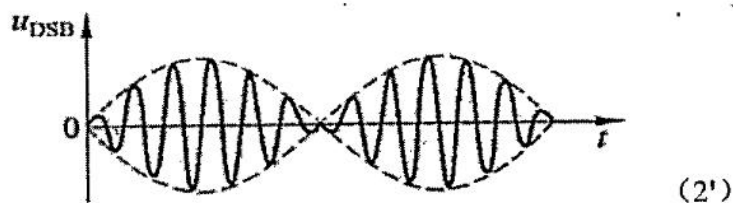
$$u(t)=1.4\cos 50\pi\times 10^3t\cos 5000\pi\times 10^3t\quad (2')$$

$$(2)\quad P_c=\frac{1}{2}\frac{U_{cm}^2}{R_L}=\frac{1}{2}\frac{2^2}{1}\text{ W}=2\text{ W}\quad (1')$$

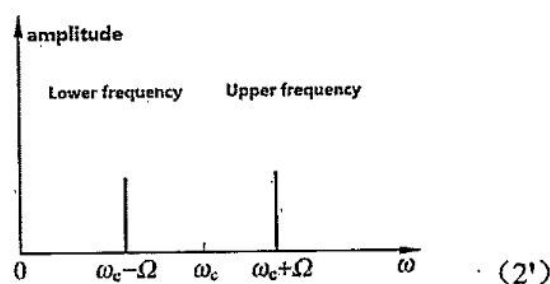
$$P_{\text{whole}}=P_{\text{sideband}}=1/2m_a^2\times P_c=0.5\times 0.7^2\times 2\text{ W}=0.49\text{ W}\quad (1')$$

$$B=2F=2\times 50\pi\times 10^3/2\pi\text{ Hz}=50\text{ kHz}\quad (2')$$

(3) 说明：能画出 DSB 的基本波形即可



说明：横坐标上也可标出具体数值



6. (10') The demodulation circuit is show in Fig. 3 below with

$$u_i(t) = 10 \cos 2\pi \times 365 \times 10^3 t + 6 \cos 2\pi \times 10^3 t \cos 2\pi \times 365 \times 10^3 t$$

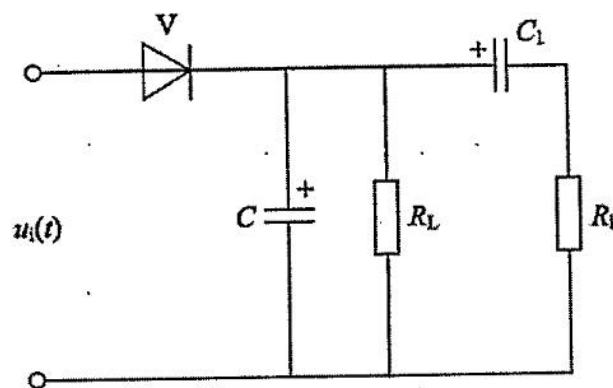


Fig. 3 demodulation circuit

The internal resistance of diode $r_D = 100 \Omega$, $C = 0.01 \mu\text{F}$ and $C_1 = 47 \mu\text{F}$. In order to ensure no waveform distortion of the output signal, please find

- (1) The maximum load resistance R_L of the demodulator;
- (2) The minimum input resistance R_i .

(1) according to the non-distortion condition, we get

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\omega C} = \frac{0.8}{0.6} \times \frac{1}{2\pi \times 10^3 \times 0.01 \times 10^{-6}} = 21.2 \text{ k}\Omega \quad (5')$$

(2) according to the non-clipping distortion condition, we have

$$m_a \leq \frac{R_i}{R_i + R_L}$$

Put $R_L = 21.2 \text{ k}\Omega$ and $m_a = 0.6$ into formula above, we get $R_i \geq 31.8 \text{ k}\Omega \quad (5')$

(注: 写出公式给 4 分, 答案对给 1 分)

7. (9') Known that the modulation frequency $f=2$ kHz, $U_a=2.5$ V, the carrier signal $u_c(t)=10\cos 2\pi \times 10^6 t$ (V), and maximum frequency shift $\Delta f_{\max}=100$ kHz, please:

- (1) Calculate the maximum phase shift m_f .
- (2) Write down the mathematical expression of the output FM signal $u_{FM}(t)$.
- (3) Calculate the effective bandwidth B_{FM} .

$$(1) \quad m_f = \Delta f_{\max} / f = 100 \times 10^3 \text{ Hz} / 2000 \text{ Hz} = 50 \quad (3')$$

$$(2) \quad u_{FM}(t) = U_{cm} \cos(\omega_c t + m_f \sin \Omega t) = 10 \cos(2\pi \times 10^6 t + 50 \sin 4000\pi t) \quad (3')$$

$$(3) \quad B_{FM} = 2(m_f + 1)f = 2 \times (50 + 1) \times 2000 \text{ Hz} = 204 \text{ kHz} \quad (3')$$